

AI-Powered Mining Equipment Monitoring and Failure Prediction System

A digital-twin based predictive maintenance platform for heavy mining equipment

Technical and Commercial Report · Working prototype · Fully open-source technology stack
Semi-finalist, TEKNOFEST 2026 Mining Technologies Competition

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github.com/krmnlhmza/AI-Powered-Mining-Equipment-Monitoring-and-Failure-Prediction-System

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Executive Summary

In underground mining, heavy equipment is the backbone of the operation. Yet the health of these machines is still managed largely reactively: intervention happens only after a failure occurs. This carries three concrete costs — high maintenance expenditure, sensor data that is lost before it is ever analysed, and fatal accidents caused by equipment failure.

The system developed by ÇankaYazılım is an integrated predictive maintenance platform that continuously monitors heavy equipment on a digital twin, predicts failures before they occur, and delivers the corrective procedure to the operator instantly. What separates it from the passive monitoring dashboards on the market is that it does not merely display data: it converts data into a decision and triggers an automated action chain (maintenance work order, report, notification).

The platform rests on a hybrid architecture in which four AI components work together: Isolation Forest for real-time anomaly detection, an LSTM network for remaining useful life (RUL) prediction, a RAG-based technical assistant that retrieves solutions from service documentation, and an n8n automation layer that starts the notification and reporting chain on critical events. The entire technology stack is built from open-source components: there is no software licence cost, and because the system runs fully on-premise, data never leaves the operator's own infrastructure.

The prototype is operational end to end. In validation testing, 2,250 sensor readings produced under normal operating conditions generated no false alarms, and all 480 fault events created across eight distinct failure types were detected.

0.00%False alarm rate under
normal operation**100%**Detection rate across
fault events**\$0**Software licence
cost**8**Validated failure
scenarios

Who is this report for?

Mining operators and their maintenance / occupational-safety functions, heavy equipment service providers, industrial digital-transformation and Industry 4.0 investors, and organisations seeking technology partnerships.

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Problem Definition

Underground mining operations must digitalise rapidly to sustain productivity against deepening reserves, declining ore grades and harsh working conditions. The SCADA panels deployed in the field only display sensor data on a screen; because micro-level vibration and current shifts invisible to the human eye are never analysed, approaching failures cannot be seen in advance. The problem manifests along three dimensions.

2.1 Maintenance cost

In heavy industrial operations such as mining, maintenance expenditure can reach up to 60% of total production cost (Savolainen & Urbani, 2021). Under a reactive model the repair itself is often short, but late detection and spare-part waiting times stretch production stoppages into hours or even days, making the operation far more expensive than planned maintenance would.

2.2 Wasted data

An autonomous mining machine generates roughly 2.5 TB of data per day through approximately 180 sensors, yet only about 1% of that data is actually used in decision-making (Don et al., 2025). The remainder is lost unanalysed because systems cannot communicate with one another. What is missing in the field is not sensors or data, but the AI layer that turns that data into a timely, meaningful decision.

2.3 Occupational safety

In underground mining, more than 40% of the most severe accidents result from being struck by or caught in a machine, and 43% of fatalities are associated with heavy haulage equipment such as loaders and trucks (CDC/NIOSH). An equipment failure is therefore not only an economic problem but a life-safety problem, and it must be caught before it becomes a physical accident.

Real case — MSHA Final Report, 22 February 2021

In an underground mine, the brakes of a locomotive whose brake shoes had worn below the manufacturer's limit and whose maintenance had been neglected failed to hold, and a 26-year-old operator lost his life. The official report identified the root cause as "operating with a defective brake system and aggravated conduct". The case demonstrates that unmonitored equipment wear is not merely costly but lethal — and why predictive maintenance is critical.

3 Literature and State of the Art

More than twelve academic studies on anomaly detection, remaining useful life prediction, RAG and digital twins were reviewed during this project. Each of these components is individually mature in the literature; however, no study combines all four in a single, vendor-independent, bidirectional platform. That is precisely the gap this project fills.

#	Study (Author, Year & Title)	Source	Contribution / finding
1	Savolainen, J. & Urbani, M. (2021). Maintenance optimization for a multi-unit system with digital twin simulation.	Journal of Intelligent Manufacturing, 32(7), 1953-1973.	Mining maintenance up to 60% of production cost; fleet optimisation via digital twin.
2	Don, M. G., Wanasinghe, T. R., Gosine, R. G. & Warrian, P. J. (2025). Digital Twins and Enabling Technology Applications in Mining.	IEEE Access, 13, 6945-6963.	Digital twin trends in mining; a machine generates 2.5 TB/day, only 1% is used.
3	van Eyk, L. & Heyns, P. S. (2025). A framework to define, design and construct digital twins in the mining industry.	Computers & Industrial Engineering, 200, 110805.	Without bidirectional data flow a system cannot be considered a true 'twin'.
4	Kuş, Ş., Tatar, F. & Toprakal, E. (2023). Digital Twin in Rail Systems.	Orclever Proceedings of Research and Development, 3(1), 104-114.	Digital twin (BIM) application in rail systems and resulting cost savings.
5	Liu, F. T., Ting, K. M. & Zhou, Z.-H. (2008). Isolation Forest.	2008 Eighth IEEE Int. Conf. on Data Mining (ICDM), 413-422.	Isolation Forest — the foundational anomaly detection algorithm.
6	Hochreiter, S. & Schmidhuber, J. (1997). Long Short-Term Memory.	Neural Computation, 9(8), 1735-1780.	LSTM — the base architecture for time series and RUL prediction.
7	Neupane, D., Bouadjene, M. R., Dazeley, R. & Aryal, S. (2025). Data-driven machinery fault diagnosis: A comprehensive review.	Neurocomputing, 627, 129588.	Comprehensive map of data-driven fault diagnosis.
8	Muratbakeev, E., Kozhubaev, Y., Novak, D., Ershov, R. & Wei, Z. (2025). Monitoring and Diagnostics of Mining Electromechanical Equipment Based on Machine Learning.	Symmetry, 17(9), 1548.	Machine-learning diagnostics applied directly to mining equipment.
9	Bharatheedasan, K., Maity, T., Kumaraswamidhas, L. A. & Durairaj, M. (2025). Enhanced fault diagnosis and RUL prediction of rolling bearings using a hybrid MLP-LSTM network model.	Alexandria Engineering Journal, 115, 355-369.	High performance of hybrid approaches such as ours.

#	Study (Author, Year & Title)	Source	Contribution / finding
10	Lewis, P., Perez, E., Piktus, A., Petroni, F. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.	Advances in Neural Information Processing Systems (NeurIPS), 33, 9459-9474.	RAG — the foundational paper behind our technical assistant layer.
11	Chen, L.-C., Pardeshi, M. S., Liao, Y.-X. & Pai, K.-C. (2025). Application of retrieval-augmented generation for interactive industrial knowledge management via a large language model.	Computer Standards & Interfaces, 94, 103995.	Semantic access to technical documentation via RAG in industry.
12	Cacciuttolo, C., Atencio, E., Komarizadehasl, S. & Lozano-Galant, J. A. (2024). IoT LoRaWAN-Based Wireless Sensors Network for Underground Mine Monitoring.	Sensors, 24(21), 6971.	Underground wireless sensor network (WSN) infrastructure.

Table 1. Principal studies in the literature and the gap addressed by this project.

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Solution Approach

Most systems in the sector are passive monitoring dashboards that merely project sensor data onto a screen; the literature calls these "digital shadows" (van Eyk & Heyns, 2025), because they present data in one direction without producing a decision from it. Our solution diverges exactly here: it processes data into a meaningful decision and uses that decision to start an automated action chain without human intervention. This closed decision loop — moving from monitoring to autonomous action — is what makes it a genuine digital twin rather than a passive dashboard.

4.1 Mapping problems to solution components

Problem	Our solution component	Outcome
Reactive maintenance: high cost and unplanned downtime	Isolation Forest (anomaly) + LSTM (remaining life)	Failures caught days in advance → predictive maintenance
99% of sensor data is lost without analysis	Digital twin + AI decision layer	Raw data becomes an instant, autonomous decision
Slow access to knowledge → long repair time (MTTR)	RAG-based technical assistant	Procedure retrieved from the manual within seconds
Equipment-related occupational safety risk	Anomaly / early-warning layer	Accident prevented before it becomes physical

Table 2. Mapping of the identified problems to our solution components and outcomes.

4.2 Comparison with conventional monitoring

Aspect	Conventional monitoring (digital shadow)	ÇankaYazılım (digital twin)
Data flow	One-way — only displays data on screen	Data → decision → automated action
Decision	Interpreted by humans, manual intervention	Automatic anomaly detection + criticality decision
Artificial intelligence	Fixed threshold / single model	Hybrid: Isolation Forest + LSTM
Technical support	Presents raw data or an error code	RAG assistant: instant solution from the manual
Action	Passive alarm; the operator must call	Automatic work order + PDF report + notification

Table 3. Conventional monitoring compared with our solution.

5 System Architecture

The system is built on a five-layer, microservice-based architecture reaching from the field to the user interface. All services are packaged with Docker and Docker Compose, so the platform runs portably on a local (edge) server or in the cloud. Every component can be updated or replaced independently.

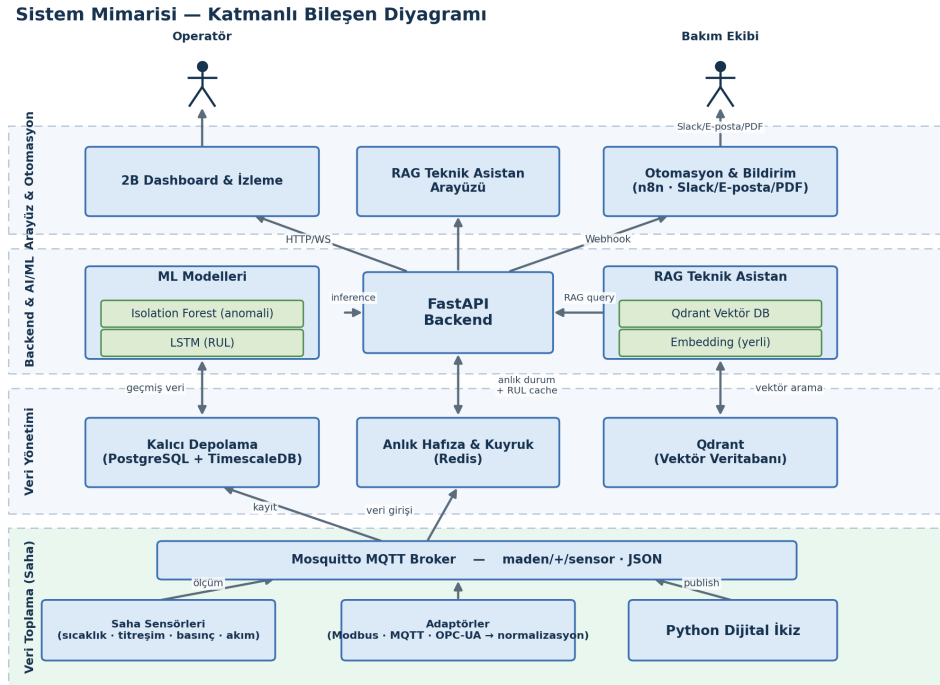


Figure 1. Five-layer microservice architecture: data acquisition, messaging, data management, backend & AI, and interface/automation layers.

- Data acquisition (field): Temperature (°C), vibration (mm/s), pressure (bar), current (A), speed (km/h), engine speed (rpm), torque (Nm) and fuel consumption (L/h) are read through the Modbus, MQTT and OPC-UA adapters common in the field, then normalised into a shared JSON schema.
- Messaging: Each normalised reading is published in real time through an Eclipse Mosquitto MQTT broker. MQTT was chosen because it consumes little bandwidth, has very low latency, and its QoS mechanism preserves message reliability even in underground conditions where connectivity drops.
- Data management: A subscriber service persists every message to TimescaleDB for time-series queries, writes the latest state to Redis for the live interface, and triggers anomaly detection. Vector representations of technical documents are held in Qdrant.
- Backend and AI: An asynchronous FastAPI backend exposes the anomaly, RUL and RAG services over REST endpoints.
- Interface and automation: A browser-based real-time monitoring dashboard, a component-level 2D digital twin view, the technical assistant interface, and an n8n automation that starts the notification chain on critical anomalies.

5.1 Technology stack

Layer	Component	Role
Runtime	Python 3.12, FastAPI, Uvicorn	Asynchronous REST services
Time series	TimescaleDB (PostgreSQL extension)	Persistent sensor history, fast temporal queries
Cache	Redis	Live latest state; keeps the UI off the disk

Layer	Component	Role
Vector database	Qdrant	Semantic search over service documentation
Messaging	Eclipse Mosquitto (MQTT 3.1.1)	Real-time field-to-system data path
Machine learning	scikit-learn, PyTorch	Isolation Forest, LSTM
Automation	n8n	Notification, PDF report and logging chain
Packaging	Docker, Docker Compose	Portable, single-command deployment

Table 4. Technology stack. All components are open-source or source-available (fair-code) licensed; internal use carries no software licence fee.

6 The Artificial Intelligence Layer

6.1 Isolation Forest — real-time anomaly detection

Each multivariate sensor reading (temperature, vibration, pressure, current and speed) is evaluated by an unsupervised Isolation Forest. The algorithm partitions the data at random to build a large number of decision trees; normal points descend deep into the trees, whereas points that behave differently are isolated in far fewer steps. The model learns each machine's normal operating characteristic in advance and catches deviations from that profile in real time.

- Every machine has its own model; what is 'normal' for a loader is not normal for a haul truck.
- The raw score is calibrated to a 0–1 range using a held-out validation set. A healthy machine typically stays in the 0.10–0.30 band.
- Above 0.60 the interface switches to a warning state; critical anomalies above 0.70 trigger the automation chain.
- A consecutive-confirmation rule provides stability: at least three consecutive readings must exceed the threshold before an anomaly is declared. Isolated noise spikes are filtered out, while genuine failures are confirmed within seconds.

6.2 LSTM — remaining useful life (RUL) prediction

To forecast when equipment will fail, a deep-learning LSTM regressor was trained. The model takes a time window of the last 20 sensor readings (5 features × 20 steps) as input and, by learning degradation trends in historical telemetry, produces a normalised health / remaining-life score. This score is scaled into hours and converted into a decision by thresholds: below 24 hours a planned maintenance warning is raised, and below 8 hours an urgent maintenance work order is created.

The model was trained on hundreds of run-to-failure degradation curves. A threshold-aware hybrid correction is also applied: when a sensor approaches its critical limit, the smaller of a physics-based ceiling and the model estimate is taken. The model therefore stays honest on a healthy machine while remaining life shortens realistically when a failure develops abruptly.

6.3 RAG-based technical assistant

So that the operator can obtain fast and reliable technical support at the moment of failure, manufacturer maintenance manuals are split into meaningful chunks, converted into numerical vectors by an embedding model, and stored in the Qdrant vector database. When the operator asks a question in natural language, that question is embedded with the same model and the semantically closest document sections are retrieved by cosine similarity and presented as the corrective procedure. This reduces dependence on external technical service and shortens mean time to repair (MTTR).

Accuracy principle — no hallucination

For questions that fall below the similarity threshold the system does not invent an answer; it returns "no reliable match found" and states its coverage explicitly. In industrial maintenance a wrong instruction is more dangerous than no answer. The system only produces responses grounded in source documentation.

The embedding model runs entirely locally; queries and documents are never sent to an external service. The current release uses an open-source multilingual embedding model. The roadmap includes migrating to newer open-source models with higher retrieval performance; that migration likewise incurs no additional licence cost.

6.4 Autonomous notification and reporting with n8n

When a critical anomaly is detected, the system posts to the webhook of the n8n automation platform. n8n filters events scoring above 0.70 and, without human intervention, starts three parallel jobs: an instant notification to the maintenance team (e-mail/Slack), generation of a PDF report for the event, and a system log entry. The time between detection and the maintenance team being informed is reduced to seconds.

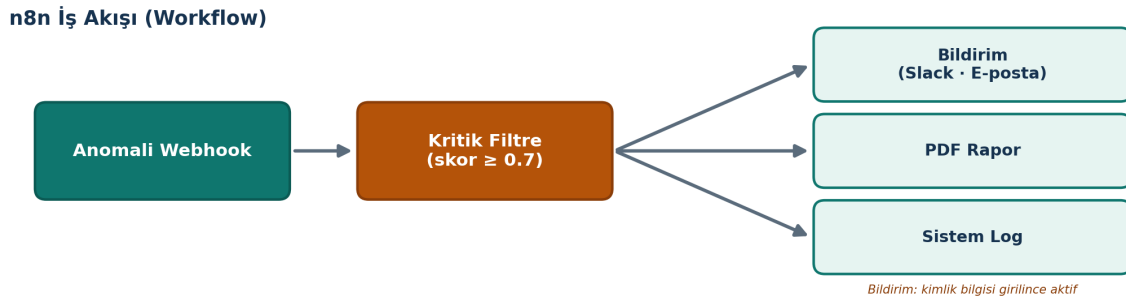


Figure 2. Autonomous notification and reporting workflow: critical anomalies are filtered, then notification, PDF report and logging jobs are triggered in parallel.

7 Data Layer and the Digital Twin Simulator

Real sensor telemetry from underground mining machines is treated as proprietary by manufacturers and is not shared externally. For development and validation we therefore built a digital twin simulator that models the physics of the machine. The simulator does not generate random numbers: it is anchored to the real operating ranges published in the manufacturer's technical documentation and preserves the physical relationships between sensors.

- **Equipment profile:** Each machine's sensor ranges are taken from the manufacturer's technical documentation (e.g. for an underground loader: temperature 72–88 °C, engine speed 800–2100 rpm, hydraulic pressure 250–280 bar, current 150–195 A).
- **Operating cycle:** The machine moves through a realistic duty cycle — idling, approaching the pile, bucket loading, loaded hauling, dumping, returning empty. Conditions such as rough terrain or loaded uphill travel can also be selected manually.
- **Physical correlation:** Sensors are not independent. As load rises, temperature, current and vibration rise together; as engine speed increases, vibration and fuel consumption increase; on a downhill run the engine brake engages, torque drops and the machine cools; hydraulic actions spike the pressure.
- **Wear and noise:** Wear accumulates with operating hours, and worn bearings shift the vibration and temperature baselines upward. Controlled Gaussian noise mimicking real sensor behaviour is added to every measurement.

The simulator feeds exactly the same path as a real field installation: generated data is published over MQTT and the rest of the system processes it as if it came from a physical sensor. Moving to the field therefore requires no architectural change — the simulator is simply replaced by the real sensor path.

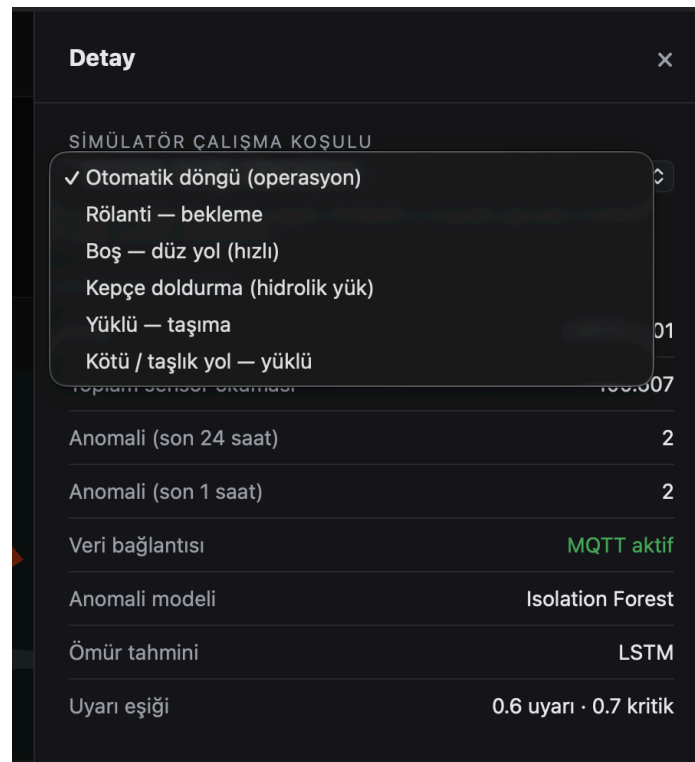


Figure 3. Physical operating-condition scenarios of the digital twin simulator.

8 Validation and Test Results

The system was measured both for its tendency to raise false alarms and for its ability to catch genuine failures. The following results come from automated tests run across three machines and all operating conditions.

Test	Scope	Result
False alarms under normal operation	3 machines × all operating conditions, 2,250 sensor readings in total	0 false alarms (0.00%)
Fault detection (event-based)	8 fault types × 3 machines × 20 repetitions = 480 fault events	All 480 events detected (100%)
End-to-end pipeline	Simulator → MQTT → database → anomaly → RUL → assistant	98,000+ readings processed without interruption
Detection latency	3-second publish interval, three-consecutive-reading confirmation	~10 seconds from fault onset to notification

Table 5. Validation tests and results.

Transparency note

These results were obtained on physically grounded simulation data derived from manufacturer specifications; they are not real field data. A production deployment requires the models to be retrained and re-validated on data collected from the site. The architecture is designed so that this retraining can be performed with a single command.

8.1 Validated failure scenarios

Eight fault types were modelled so that they produce simultaneous, physically consistent signatures across multiple sensors. In each scenario the system detects the anomaly, predicts the remaining life and directs the user to the relevant corrective documentation through the technical assistant.

Failure scenario	Sensor signature	Dominant indicator
Oil / hydraulic pump failure	Pressure drops, vibration rises, oil temperature falls, fuel rises	Pressure (low)
Bearing wear	Vibration rises markedly, temperature rises slightly	Vibration
Engine overheating	Oil temperature reaches the critical threshold, current rises	Temperature
Injector / combustion fault	Misfire vibration, fuel rises, torque drops	Vibration
Overcurrent (motor winding)	Current persistently above the upper limit, winding heats	Current
Brake overheating	Temperature rises, speed drops, vibration rises	Temperature (brake)
Transmission failure	Transmission vibration and oil temperature rise	Vibration
Cooling system failure	Engine temperature rises continuously, current stays normal	Temperature

Table 6. Validated failure scenarios and their multi-sensor signatures.

9 User Interface

The interface is a lightweight, real-time monitoring dashboard that runs in the browser and requires no additional installation. It presents information to operators and maintenance teams at three levels: fleet-level status, machine-level live telemetry and component-level failure indication.

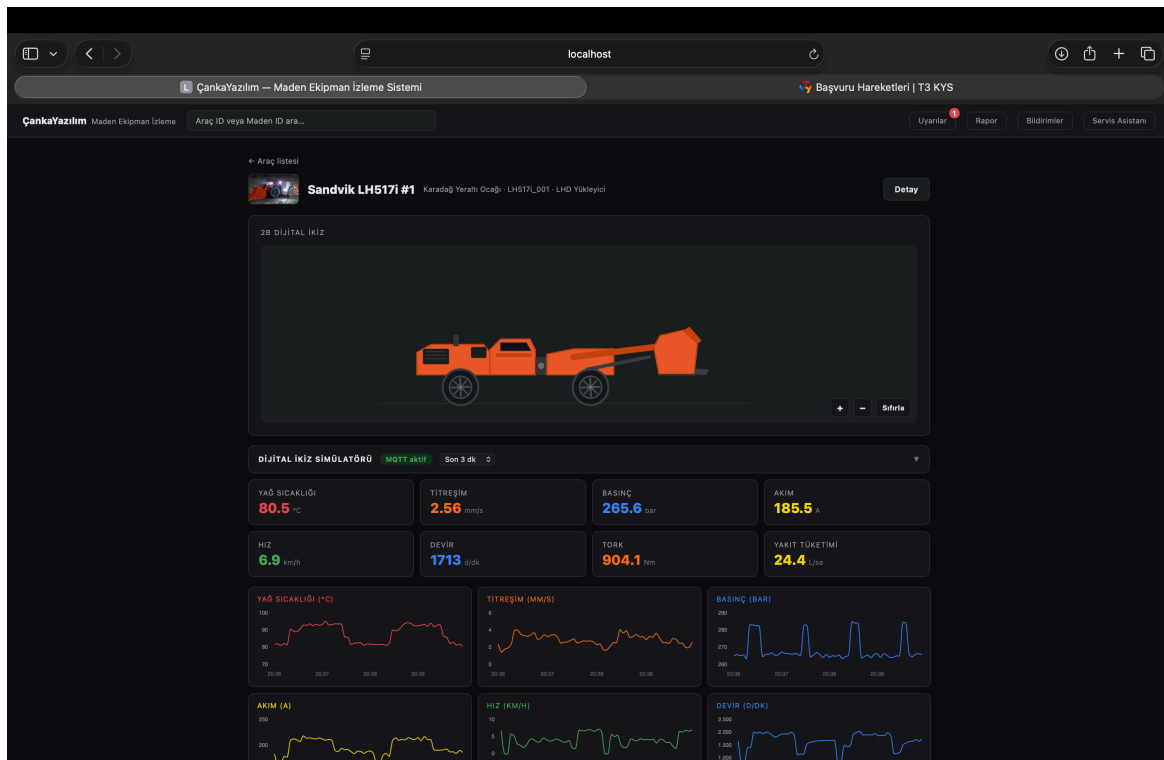


Figure 4. Live monitoring dashboard — real-time sensor cards and time-series charts. Anomaly moments are marked on the chart of the affected sensor.

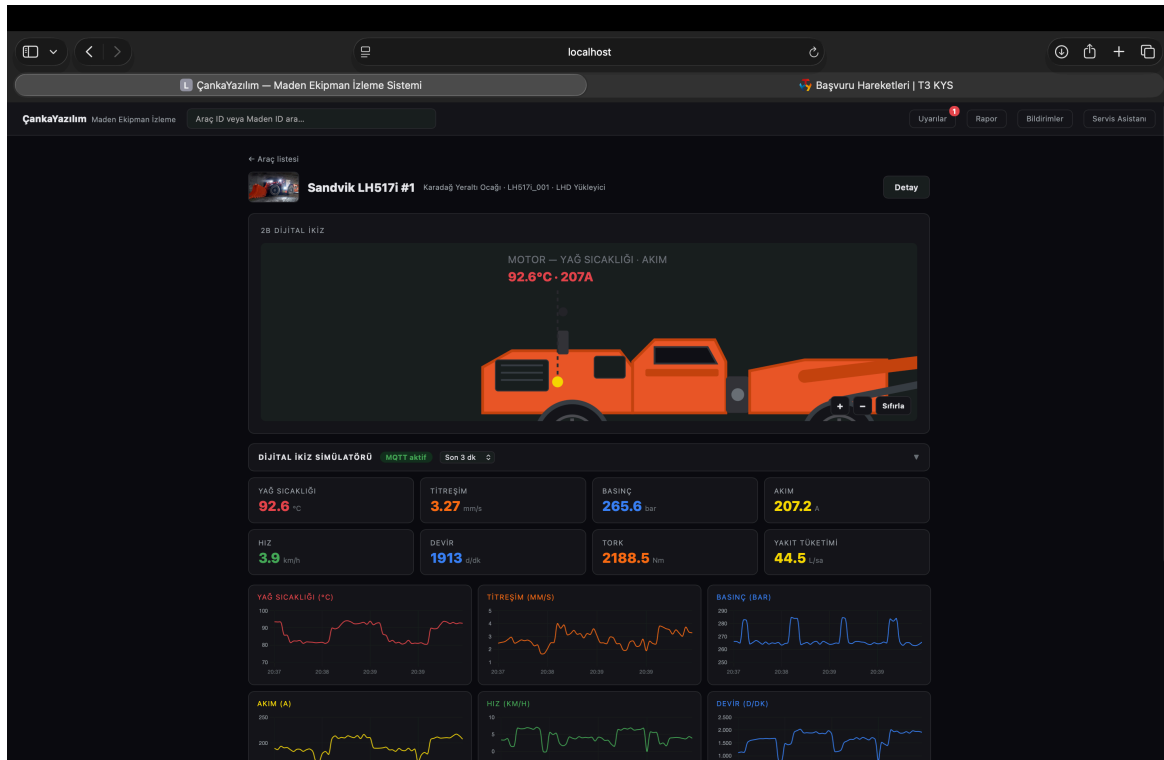


Figure 5. 2D digital twin — level of detail increases as the view is zoomed; the faulty component is highlighted on the machine schematic.

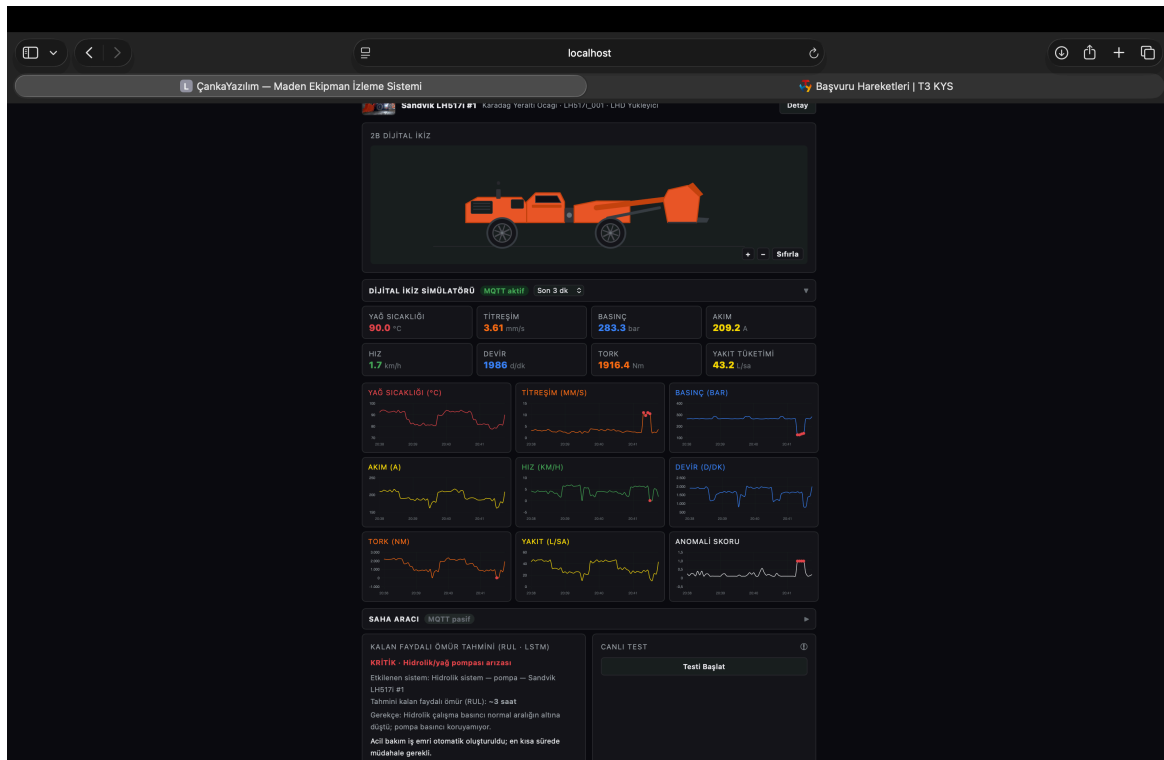


Figure 6. Anomaly detection and the LSTM-generated remaining useful life (RUL) estimate: the failure, the affected system, estimated remaining life and the recommended action.

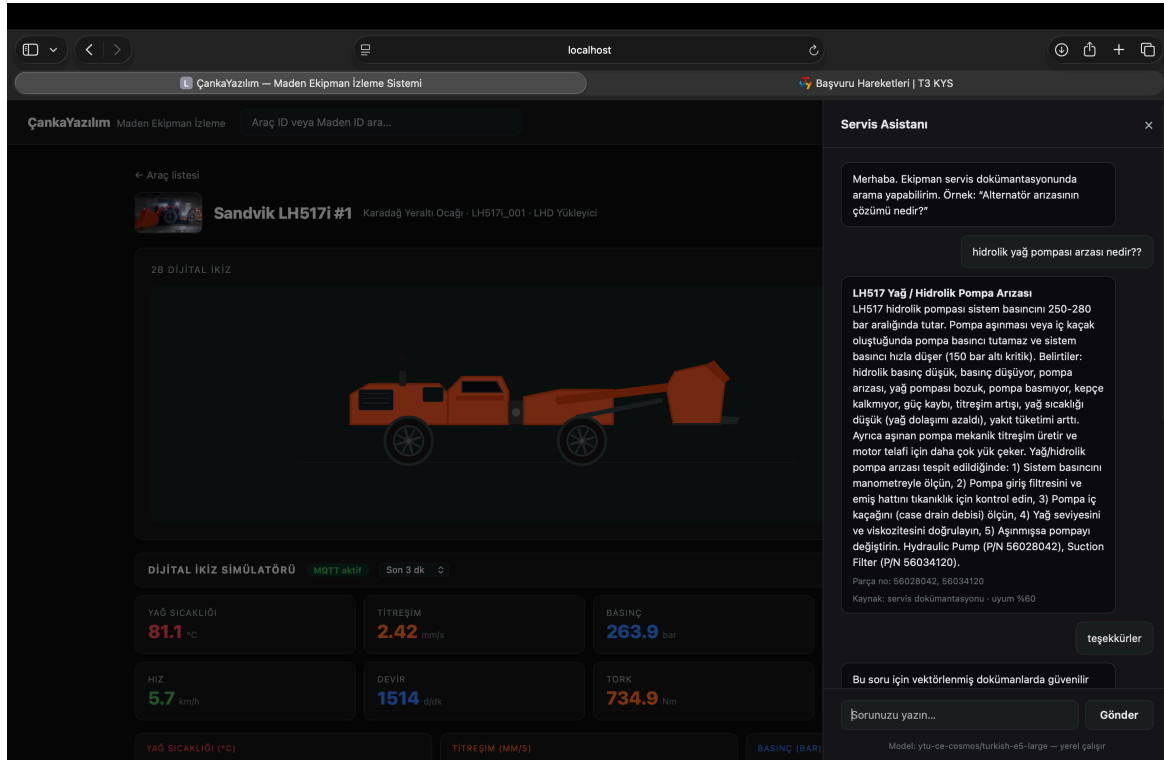


Figure 7. RAG-based service assistant — a natural-language question and the answer produced from service documentation, including the part number.

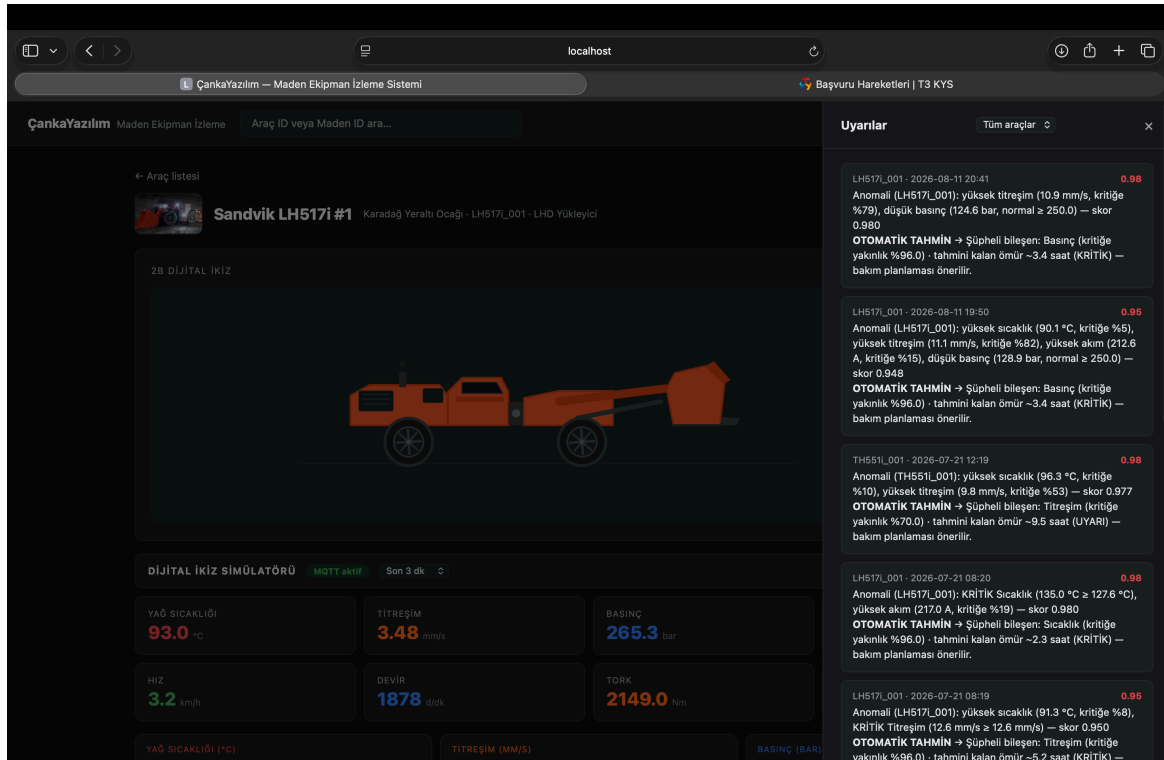


Figure 8. Alerts panel — detected anomalies, their scores and the automatically generated prediction.

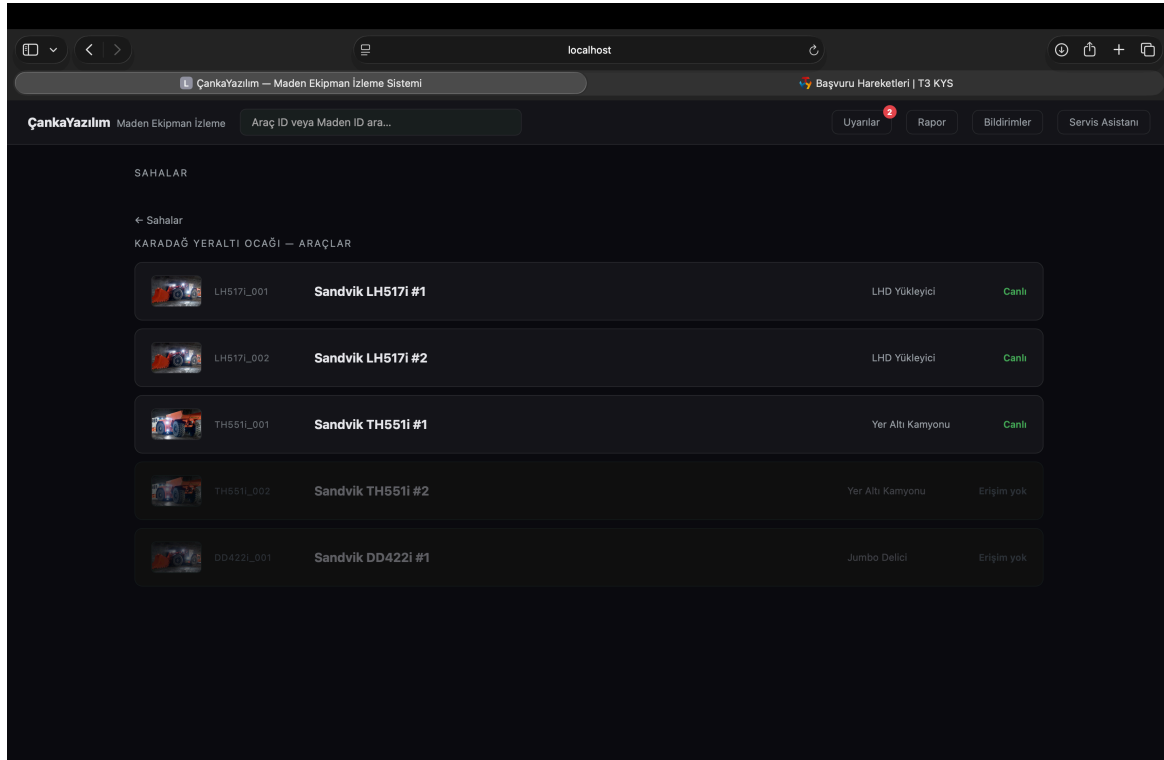


Figure 9. Site and machine selection screen — multi-site, multi-machine, vendor-independent fleet monitoring.

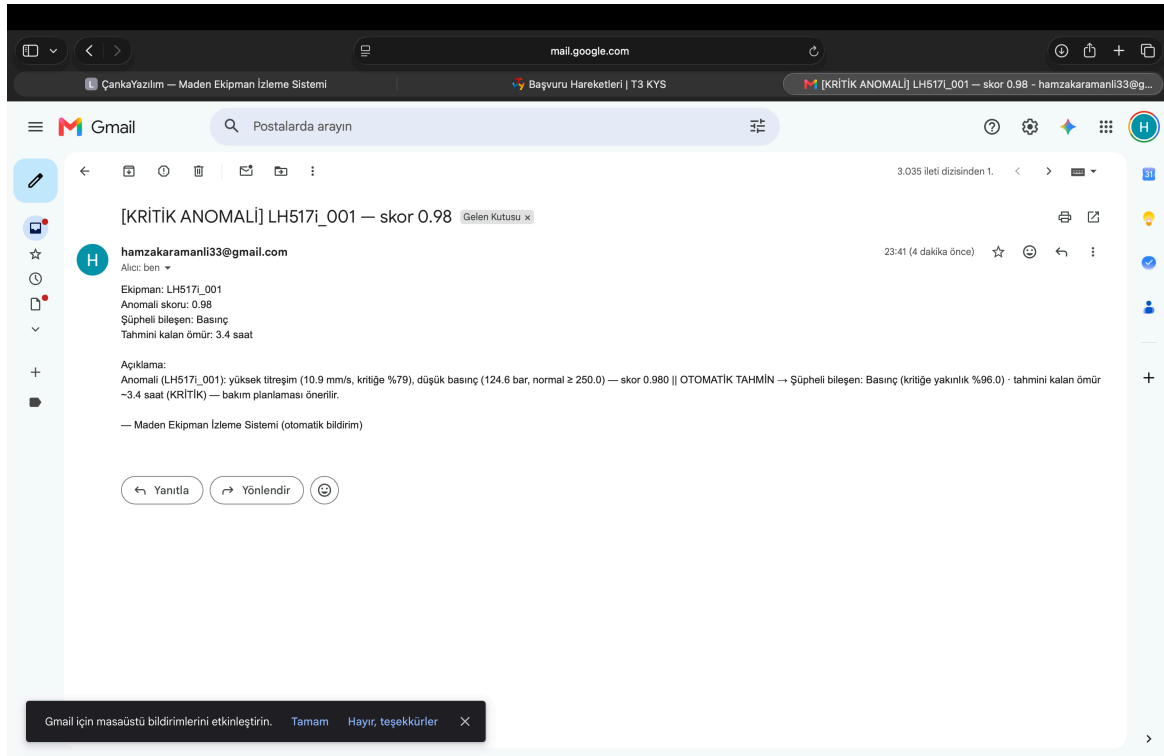


Figure 10. A real e-mail notification dispatched by the automation chain on a critical anomaly.

10 Competitive Comparison

The major international solutions (Sandvik OptiMine, Cat MineStar Health, Komatsu, Epiroc) are powerful, but they largely operate only on their own branded equipment, and none of them provides a semantic technical assistant that delivers an instant solution from the service manual to the operator. General-purpose maintenance management (CMMS) software, in turn, is not specific to mining or to failure prediction.

Capability	ÇankaYazılım	Sandvik Opti	Cat MineStar	Komatsu	Epiroc	Generic CMMS
Real-time monitoring	Full	Full	Full	Full	Full	Full
Anomaly detection	Full	Full	Full	Partial	Partial	Partial
Remaining life / failure prediction	Full	Partial	Full	Partial	Partial	Partial
RAG technical assistant (manual-grounded)	Full	None	None	None	None	None
Vendor-independent (mixed fleet)	Full	None	None	None	None	Partial
Fully on-premise / data sovereignty	Full	None	None	None	None	Partial
Open source / zero licence cost	Full	None	None	None	None	Partial
Equipment-driven safety early warning	Full	Partial	Partial	Partial	Partial	None

Table 7. Feature-level comparison of our solution against existing commercial offerings.

10.1 Four differentiators

- **Vendor independence:** The platform adapts to mixed-brand fleets from different manufacturers with no architectural change — only configuration and knowledge-base updates.
- **Hybrid AI:** Anomaly detection and remaining-life prediction operate together in a single decision-support flow; the literature likewise demonstrates the superiority of hybrid approaches (Bharatheedasan et al., 2025).
- **RAG-based technical assistant:** By turning manufacturer maintenance manuals into a semantic search engine, the system delivers the relevant corrective section to the operator at the moment of failure — a component that is pioneering in this sector.
- **Zero licence cost and data sovereignty:** The system is built entirely from open-source components and runs on-premise; there is no foreign cloud or licence dependency, and data never leaves the operation.

11 Cost and Benefit

11.1 Total cost of ownership

The distinguishing commercial property of the system is that it carries no licence cost on the software side. Every component used is open-source or source-available licensed and requires no fee for internal use. The items the operator must fund are limited to hardware, integration and maintenance.

Cost item	Status	Explanation
Software licence	None	Entire stack open-source / fair-code; no subscription or per-seat fee
Cloud subscription	Optional	The system can run on a single on-premise server; cloud is not required
Server hardware	Moderate	A single industrial server or edge device is sufficient

Cost item	Status	Explanation
Sensors / retrofit	Variable	Most data already exists on the machine; extra sensors only for missing measurements
Integration and setup	One-off	Connects to existing infrastructure over Modbus/OPC-UA/MQTT
Maintenance and updates	Low	Microservice structure allows components to be updated independently

Table 8. Total cost of ownership items.

11.2 Expected benefit

The impact of predictive maintenance is measured in the literature: a 30–50% reduction in unplanned downtime and an 18–25% reduction in maintenance cost (McKinsey, 2020). A domestic digital twin study for rail systems likewise projected approximately 12–25% savings in maintenance cost (Kuş et al., 2023). In a sector where maintenance can reach up to 60% of total production cost, these figures translate directly into profit.

30-50% Potential reduction in unplanned downtime	18-25% Potential reduction in maintenance cost	MTTR ↓ Repair time shortened by the technical assistant	Safety ↑ Failures caught before they become accidents
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Beyond the economic benefit there are two further contributions. On occupational health and safety, sudden brake, hydraulic or engine failures are caught before they turn into a physical accident, and emergency intervention in confined, hazardous environments is replaced by planned maintenance under controlled conditions. On the environmental side, motor current analysis supports energy efficiency, while avoiding unnecessary part replacement reduces waste generation and the carbon footprint.

Sürdürülebilirlik: Finansal · Çevresel · Sosyal



Figure 11. Sustainability dimensions: financial, environmental and social/safety contributions.

12 Feasibility and Roadmap

The methods used (Isolation Forest, LSTM, vector-based RAG) are mature, validated algorithms in both industry and academia, which lowers the technical risk of the solution. The system is designed to operate across different equipment brands and models without architectural change. Telemetry from existing PLC/SCADA infrastructure can be integrated over the Modbus and MQTT industrial protocols.

Ticarileşme ve Olgunluk Yol Haritası

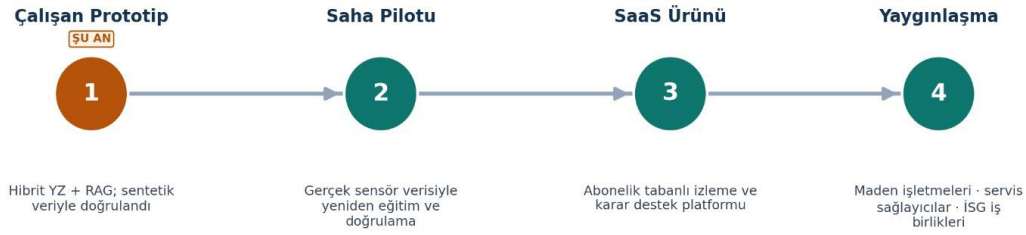


Figure 12. Maturity and commercialisation roadmap: working prototype, field pilot, productisation and scale-up phases.

12.1 Next steps

Phase	Scope	Output
Field pilot	Real sensor integration on 2-3 machines at a single operation; retraining the models on site data	Model performance validated on real data
Service manual ingestion	Full ingestion of manufacturer maintenance manuals into the vector database	A technical assistant covering thousands of pages
Advanced embedding model	Migration to newer open-source embedding models with higher retrieval performance	More accurate document retrieval, still at zero licence cost
3D digital twin	Three-dimensional, component-level visualisation of the failure	A clearer failure representation for the operator
Productisation	Multi-tenant structure, role-based access control, monitoring at scale	A product deployable across multiple operations

Table 9. Commercialisation roadmap and next steps.

12.2 Risks and mitigation

Risk	Mitigation approach
Difficulty accessing real field data	Development on a physics-based simulator; pilot partnerships; gradual retraining on data collected from the site
Resistance to adopting new software	Parallel deployment over standard protocols without touching existing SCADA/PLC infrastructure
Cyber security of critical infrastructure	Fully on-premise deployment; no external service dependency; data never leaves the operation
Risk of model false alarms	Consecutive-confirmation rule and per-machine calibration; measured false alarm rate of 0.00%

Table 10. Principal risks and mitigation approaches.

13 Project Team and Contact

The project is carried out under the name ÇankaYazılım. The system architecture, AI layer, backend services, data infrastructure and user interface were developed end to end and validated in working condition. The project reached the semi-final of the TEKNOFEST 2026 Mining Technologies Competition.

Contact

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Source code and technical documentation:
github.com/krmnlhmza/AI-Powered-Mining-Equipment-Monitoring-and-Failure-Prediction-System

For pilot deployments, technology partnerships or a detailed technical evaluation, please get in touch at the addresses above. A live demonstration of the system can be provided on request.

14 References

The principal sources cited in this report are listed with full bibliographic details in Table 1. The following institutional sources were also used:

- McKinsey & Company (2020). Analytics-driven maintenance technologies. — Figures on the effect of predictive maintenance on downtime and cost.
- CDC / NIOSH (2024). Machinery- and haulage-related mining fatalities. — Equipment-related fatality rates in mining accidents.
- MSHA — Mine Safety and Health Administration (2021). Final accident report, 22 February 2021. — Fatal accident caused by brake failure.
- Sandvik technical product documentation — Real equipment operating ranges used in the simulator.